2025-Winter Paper 1 – Basic Sciences in Urology

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Q1. Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management.

Type: LAQ (25 marks) (related: 2021-Summer Paper 1, Q3e; 2016-Summer Paper 3, Q2)

Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

Pathogenesis

Type	Lesion	Typical patient
Intrinsic (commonest)	Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly	Neonate or infant, antenatal finding
Extrinsic	**Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it	Older child or adult with intermittent pain
Secondary	Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion	Any age

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.
- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

Diagnosis

1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

2. Postnatal ultrasound after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

4. Others: VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm H_2O obstructed) if equivocal.

Management

Conservative: most resolve, about **one-third need surgery**; serial ultrasound and renogram.

Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.
- Bilateral or solitary kidney.

Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.
- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

Complications and follow-up

- Urine leak, stent problems, UTI; early failure from oedema or clot (temporary nephrostomy).
- **Late failure:** redo pyeloplasty (best), endopyelotomy or ureterocalicostomy. Renogram and ultrasound after stent removal.

Recent advances

- Robotic redo with buccal graft and ICG (indocyanine green) fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

> **Key points to remember:**
>
> - Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
> - SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
> - Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
> - Anderson–Hynes ~95%; never divide a crossing vessel.

Q2. Nepron Sparing Surgery in renal malignancy.

Type: LAQ (25 marks) (also answered in 2022-Summer Paper 3, Q1; 2017-Summer Paper 1, Q1)

"Nepron" is read as nephron.

Definition and rationale

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why:** RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to cardiovascular events and death. PN gives equal cancer control for small masses and keeps about **90%** of global function (RN 65–70%).

Indications

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney; RN would mean dialysis
Relative	Contralateral kidney threatened: stones, renal artery stenosis, hydronephrosis, diabetes, severe hypertension; hereditary RCC
Elective	**cT1a (≤4 cm)** with a normal contralateral kidney: **standard of care**

- Also used for selected cT1b–T2, complex or hilar tumours when function matters.
- **AUA (American Urological Association) 2021:** RN is preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new eGFR (estimated glomerular filtration rate) likely **>45** after RN. Otherwise PN.
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1.

Preoperative evaluation

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR and proteinuria.
- **R.E.N.A.L. nephrometry** (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines): **4–6 low, 7–9 moderate, 10–12 high** complexity. PADUA and C-index are alternatives.
- Biopsy when it changes management; genetic counselling if age ≤46, multifocal or bilateral.

Techniques of PN

Approach: open (flank extraperitoneal or transperitoneal), laparoscopic, or **RAPN (robot-assisted PN)**: transperitoneal for anterior and retroperitoneal for posterior tumours.

Steps

1. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
2. Defat the kidney; **intraoperative ultrasound** maps borders and depth.
3. Clamp the artery (± vein). Use **ice slush** if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
4. Excise with a thin rim of normal parenchyma. **A negative margin matters; its width does not.**
5. **Two-layer renorrhaphy:** inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.

6. Unclamp; check haemostasis; close Gerota fascia; drain.

Excision options

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma, thin rim	Along the pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma
Use	Default	Hereditary RCC (VHL [von Hippel–Lindau]), multifocal, severe CKD

- Polar resection or heminephrectomy for large polar tumours.

Vascular control (ischaemia)

- Traditional warm ischaemia **<25–30 minutes**. Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Off-clamp, selective or early unclamping:** randomised trials show no long-term eGFR advantage. Hypothermia and mannitol give no benefit with a normal contralateral kidney.

Ablative NSS

- **Cryoablation** (double freeze–thaw), **RFA (radiofrequency ablation)**, **MWA (microwave ablation)**; mostly percutaneous.
- AUA: alternative for **cT1a <3 cm**, especially elderly or comorbid; biopsy before or at ablation. Recurrence is higher than PN but treatable by repeat ablation.
- **Also know:** SBRT (stereotactic body radiotherapy) for non-surgical candidates; active surveillance for frail patients.

Special situations

- **Hereditary RCC:** observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision).
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.
- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** neoadjuvant TKI (tyrosine kinase inhibitor) downsizing in imperative cases; bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

Outcomes and evidence

- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.
- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** PN vs RN for ≤5 cm: better function and similar oncology with PN; 10-year overall survival 81% RN vs 76% PN, but the trial is flawed.

Complications

- **Urine leak** (drain, stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (delayed haematuria: angioembolisation); acute kidney injury.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** if >50% of a solitary kidney is removed: proteinuria, FSGS (focal segmental glomerulosclerosis); ACE (angiotensin-converting enzyme) inhibitor.
- **Follow-up:** AUA risk-based imaging; eGFR and urine protein at every visit.

Recent advances

- 3D models (randomised: shorter ischaemia, less blood loss); ICG (indocyanine green) fluorescence for selective clamping; single-port RAPN.

Diagram to draw: clamped renal artery, tumour excised with a rim, two-layer renorrhaphy.

Key points to remember:

- Imperative, relative, elective; PN is standard for cT1a.
- AUA: RN only if complex + no CKD + eGFR likely >45.
- Preserved parenchyma, not clamp time, governs function.
- EORTC 30904: better function, similar cancer control.

Q3a. Extended Criteria for Selection of Live Renal Donor

Type: SAQ (10 marks) (donor assessment also answered in 2022-Summer Paper 3, Q3a; marginal donor in 2024-Winter Paper 4, Q3e)

Read as the expanded (marginal, "medically complex") living donor.

Definition and principle

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold. Donor safety comes first.
- Counsel with figures: donor lifetime ESKD 90 vs 14 per 10,000 matched non-donors (Muzaale 2014); mortality 3 per 10,000.

Extended criteria

Factor	Standard	Extended (individualised)	Exclude
Age	Adult	Older donors (often >60) acceptable; no upper limit	Under legal age
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	**60–89**	<60
Albuminuria, mg/day	<30	**30–100**	>100
Hypertension	None	Controlled on 1–2 drugs, no target-organ damage	Uncontrolled, organ damage
Obesity	BMI (body mass index) <30	**30–35**	Many centres: >35
Glucose	Normal	Prediabetes; selected type 2 diabetes	Type 1 diabetes
Stones	None	Single past stone, low recurrence risk; small stone removed ex vivo	Bilateral or recurrent; cystinuria, primary hyperoxaluria
Anatomy	Single vessels	Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia)	Major anomaly

- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype; future pregnancy (pre-eclampsia risk).

Safeguards

- Measured GFR and split function; **the better kidney stays with the donor**.
- CT (computed tomography) angiography; laparoscopic retrieval.
- India: **THOA (Transplantation of Human Organs Act)** authorisation; independent donor advocate.
- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

Diagram to draw: risk-threshold scale (accept, individualise, decline).

Key points to remember:

- KDIGO 2017: risk-based lifetime ESKD threshold.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes cannot.

Q3b. Metabolic Evaluation in Stone Disease

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 2, Q2)

Rationale

- Stones are a chronic metabolic disease: about **50% recur within 10 years**. Evaluation finds a treatable cause and directs targeted therapy, which reduces stone events.

Step 1: Screening evaluation (every first-time former; AUA [American Urological Association], EAU [European Association of Urology])

Component	What is assessed
History	GI (gastrointestinal) disease, gout, sarcoidosis, recurrent UTI (urinary tract infection), family history; diet; drugs (calcium, vitamins C and D, acetazolamide, steroids)
Blood	Creatinine, sodium, potassium, calcium, uric acid, bicarbonate; PTH (parathyroid hormone) if calcium is high
Urine	pH, crystals (hexagonal = cystine), culture
Stone analysis	Infrared spectroscopy or X-ray diffraction; uric acid, cystine and struvite point to a cause
Imaging	NCCT (non-contrast computed tomography): burden

Step 2: Extended evaluation (high-risk or recurrent formers)

- **High risk:** children, recurrent stones, solitary kidney, GI disease or bariatric surgery, hyperparathyroidism, RTA (renal tubular acidosis), metabolic syndrome, gout, family history.
- **One or two 24-hour urines** on a usual diet, ≥4 weeks after an episode, stone- and infection-free; creatinine confirms completeness.

Parameter	Target per day	Abnormality
Volume	**>2.5 L**	Low volume (commonest)
Calcium	<4 mg/kg (about 250 mg women, 300 mg men)	Hypercalciuria
Oxalate	**<40 mg**	Dietary, enteric, primary
Citrate	**>320 mg**	Hypocitraturia (RTA, diarrhoea)
Uric acid	<600 mg	Hyperuricosuria
pH	5.8–6.2	<5.5 uric acid; >6.5 calcium phosphate or infection
Sodium	<100 mmol	Raises urine calcium
Cystine	Negative screen	Cystinuria

- **Special tests:** ammonium chloride load (urine pH stays >5.5 in distal RTA); quantitative cystine.
- Fast and calcium-load subtyping of hypercalciuria is no longer advised (does not change treatment).

Follow-up

- Repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects.
- Pregnancy: defer 24-hour urine.

Diagram to draw: flowchart: first stone → screening → low risk (general advice) or high risk (24-hour urine → targeted therapy).

> **Key points to remember:**
>
> - Screening for all first-time formers; 24-hour urine for high-risk.
> - Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg.
> - Repeat 24-hour urine within 6 months of therapy.

Q3c. Hematospermia

Type: SAQ (10 marks)

Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.
- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures
Inflammation and infection	Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder or seminal vesicle
Vascular	Prostatic varices, BPH vessels, AV (arteriovenous) malformation
Systemic	Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma

Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

Who needs full work-up: age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm).
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

Management

- **Young, single episode, normal tests:** reassure.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr scope through the utricle, irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

Key points to remember:

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria.
- Exclude prostate cancer and genitourinary tuberculosis.

Q3d. Vascular Complications after Renal Transplantation in Recipient

Type: SAQ (10 marks)

Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.
- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

Main complications

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Incidence	0.2–5%	0.1–8.2%	1–23% (commonest)
Timing	First days	First week	Months (usually 3 months–2 years)
Presentation	Sudden anuria	Painful swollen graft, haematuria, oliguria	Refractory hypertension, rising creatinine, flash pulmonary oedema
Doppler	No arterial flow	**Reversed diastolic flow**, clot in vein	High velocity at stenosis; **parvus-tardus** waveform distally
Treatment	Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable	Immediate exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty or stenting** (first choice); surgical revision if that fails

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

Prevention

- Anastomose away from atheroma; check the lie for kinking.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.

- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

Diagram to draw: Doppler waveforms: normal, parvus-tardus (TRAS) and reversed diastolic flow (venous thrombosis).

> **Key points to remember:**
>
> - TRAS is the commonest; angioplasty or stent first.
> - Venous thrombosis: reversed diastolic flow, swollen tender graft.
> - Arterial thrombosis: sudden anuria; explore immediately.

Q3e. Management of Acute Hyperkalemia

Type: SAQ (10 marks)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.
- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers, suxamethonium
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	Jejunal conduit (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.
- **ECG changes:** peaked T (earliest), prolonged PR, wide QRS, loss of P, sine wave, ventricular fibrillation. Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; lasts 30–60 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; check glucose (hypoglycaemia)
	Salbutamol 20 mg nebulised	Additive to insulin
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours
	Haemodialysis	Refractory, oliguric, ESKD (end-stage kidney disease)

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or K^+ ≥6.5 → calcium → shift → remove.

Key points to remember:

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium.

Q3f. BCG therapy in Bladder Cancer

Type: SAQ (10 marks) (intravesical therapy also answered in 2017-Summer Paper 4, Q1)

Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is the only intravesical agent that reduces both **recurrence and progression** of NMIBC (non-muscle-invasive bladder cancer).

Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced frequency was inferior.

Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

Toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Symptomatic
2	Severe or >48 h	Isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; add ethambutol; steroids only with cover

- BCG is resistant to **pyrazinamide**.

BCG failure

- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite **adequate BCG** (≥5 of 6 induction + ≥2 of 3 maintenance), or HG relapse within 6 months (Ta/T1) or 12 months (CIS). **Timely radical cystectomy** is standard.

- **If unfit or unwilling:** pembrolizumab, nadofaragene firadenovec, N-803 + BCG, gemcitabine–docetaxel.
- **Recent:** PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC); TAR-200 gemcitabine-releasing device.

Diagram to draw: SWOG schedule timeline.

> **Key points to remember:**
> - Only BCG reduces progression; maintenance is essential.
> - Never immediately after TURBT; start at 2–4 weeks.
> - Unresponsive: timely cystectomy.

Rapid Revision Sheet – 2025-Winter Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. (LAQ)

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- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.

Q2. Nepron Sparing Surgery in renal malignancy. (LAQ)

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- AUA: RN only if complex + no CKD + eGFR likely >45.
- Preserved parenchyma, not clamp time, governs function.
- EORTC 30904: better function, similar cancer control.

Q3a. Extended Criteria for Selection of Live Renal Donor (SAQ)

- KDIGO 2017: risk-based lifetime ESKD threshold.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes cannot.

Q3b. Metabolic Evaluation in Stone Disease (SAQ)

- Screening for all first-time formers; 24-hour urine for high-risk.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg.
- Repeat 24-hour urine within 6 months of therapy.

Q3c. Hematospermia (SAQ)

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria.
- Exclude prostate cancer and genitourinary tuberculosis.

Q3d. Vascular Complications after Renal Transplantation in Recipient (SAQ)

- TRAS is the commonest; angioplasty or stent first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Q3e. Management of Acute Hyperkalemia (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium.

Q3f. BCG therapy in Bladder Cancer (SAQ)

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Unresponsive: timely cystectomy.